

Machine yearning: LLMs do not capture formal linguistic structure and obscure neuroscientific inquiry

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[Accepted, in press: *Behavioral and Brain Sciences*]

Abstract:

Futrell and Mahowald (2025) argue that neural networks have learned non-trivial aspects of language. We argue that these systems have not in fact demonstrated “mastery” (2025: 10) of syntax, marshalling recent evidence, and that they further serve to obscure explanatory insights with respect to topics in the cognitive neuroscience of language.

Futrell and Mahowald (2025) (FM) argue that LLMs have mastered formal properties of human language. They point to work on the geometry of word embeddings and their ability to capture syntactic dependencies (Diego-Simón et al. 2024). However, dependency grammar is not a viable candidate theory of natural language syntax, since it isolates word-word dependency graphs rather than hierarchical constituency structure (Marcolli et al. 2025) (dependency graphs are projections from this deeper algebraic structure). While FM (2025: 42) argue that recent work serves to “refute the claim that neural LMs can learn any language” (i.e., Kallini et al. 2024), this work is confounded by the fact that Kallini and colleagues did not contrast count-based vs. *constituency-based non-adjacency* rules. Instead, they contrasted count-based non-adjacency with simple adjacency, which is not the appropriate resolution to isolate constituency-based rules that are not operationalized over linear distance (Hunter 2024). Other work casts doubt on the claim that LLM probabilities can distinguish between possible/impossible languages: Leivada et al. (2025a) show that LLM probabilities do not constitute reliable proxies for model-internal representations of syntactic knowledge, and Yang et al. (2025) show that perplexity scores from GPT-2 small do not distinguish attested versus unattested word orders.

Natural language syntax-semantics is not exhausted in formal and conceptual scope by distributional similarity or predictive probability. Many words are able to jointly host analog/continuous and digital representations (Jackendoff & Erk 2025), alongside categorially incompatible types of inherent polysemy (Murphy 2024a) (consider what kind of thing *school* is meant to be in ‘The progressive school next to the river was high in the *NYT* rankings and had just hired a new chancellor after being repainted’). In contrast to vector space semantic models operating within a single homogeneous metric space, natural language semantics is highly typed and heterogeneous: the cognitive architecture underlying it has to coordinate multiple representational formats (spatial structure, event structure, theory of mind, quantification, Boolean logic, causation, telicity, evidentiality). These conceptual domains are not “dimensions” in the

embedding sense. They host different mathematical structures with different neural substrates (Baggio 2018, Murphy 2025).

If FM are right that LLMs have learned non-trivial aspects of language, what have they learned when they affirm that ‘Eight eighteen resigned seventeen this eight twenty-one any ten warning’ is a grammatical sentence (Leivada et al. 2025b)? The field of AI has often leaned on metaphors to circumvent the black box nature of its agents (Mitchell 2024), with ‘learning’ a clear example. The notion of emergence found in FM is another. Does any putative “mastery” of syntax entail that LLMs emulate the process of emergence as understood in biology (i.e., a behavior arising de novo as the (by-)product of a complex system)? Emergence is notoriously hard to define, but there is agreement that it entails novelty that is not predicted by lower-level generic components of the adaptive system (Krakauer et al. 2025). Human-engineered systems do not straightforwardly satisfy standard criteria for strong emergence, because they are purposely designed to emulate target behavior. In this context, anthropomorphizing labels such as *learning*, *emerging* and *reasoning* are more accurately described as wishful mnemonics (McDermott 1981) – a textbook case of machine yearning – than reasonable scientific claims documenting model capabilities.

LLMs can surely capture certain statistical regularities from the output of language (text, data), but they do not infer grammar the way humans do and fail to capture fundamental principles of linguistic structure (Murphy et al. Forthcoming). They do not reliably identify grammaticality (Dentella et al. 2024, 2025a). Instead of performing higher-order reasoning, they have been shown to engage in “sophisticated pattern matching” (Shojaee et al. 2025). They have a strong linearity bias and do not capture deep syntactic structures (Diego-Simón et al. 2025). There is no evidence that LLMs capture cross-constructural generalizations about null complementizers (Momma et al. 2025). When tested to learn restrictions on filler-gap dependencies in English and Norwegian, the generalizations learned by language models diverge from those of humans

(Kobzeva et al. 2025). This does not support the claim that filler-gap dependencies and island constraints can be learned without domain-specific biases. LLMs have a preference for manipulating noun-related information over verb-related information (Dentella et al. 2025b), thus distinguishing them from highly grammaticalized types of thought that emphasize dimensions of tense, aspect and mood. Lastly, text-to-image models also fail to represent the meaning of prompts with relatively basic syntactic structures triggering compositional meanings (Leivada et al. 2023, Murphy et al. 2025).

Within cognitive neuroscience, LLMs have arguably been unable to help with causal-mechanistic questions concerning *how* the algebraic properties of syntax-semantics are neurally enforced (Marcus 2001, Murphy 2015, 2020, 2024, 2025, Murphy et al. 2022). Many LLM-brain alignments seem to be “driven by fragile methodologies and overlooked confounds” (e.g., word rate, positional signals) (Hadidi et al. 2025). An fMRI study showed that transformers were significantly inferior at predicting neural signatures to models explicitly designed to encode the syntactic relations between words (Fodor et al. 2025). Problematic causal logic is also common in the LLM literature (e.g., affirming the consequent) (Guest & Martin 2023), whereby researchers often confuse types of inference and misunderstand the evidentiary role that correlation plays (see also Meijer 2021). A scientist or artificial model may be able to classify and predict very well – all without providing causal explanations of a system.

FM (2025: 3) warn against dismissing LLMs as approximation devices that can fit any data without revealing the true underlying system. But this analogy is apt. LLMs can mimic many linguistic phenomena because they are universal function approximators and their capacity and data exposure are unbounded. LLM fluency should not be confused with competence (Slaats & Martin 2025). We therefore stress that the functionalist approach to language sidelines the necessary contributions to theories of cognition from symbolic, algebraic structures.

98 FM invoke Kubrick's *Dr. Strangelove* in their article's title, referring to an emerging "love" for LLMs
99 winning over irrational fear and anxiety. We feel that another Kubrick movie, *2001: A Space*
100 *Odyssey*, teaches an altogether simpler and more appropriate lesson: to hesitate before trusting
101 HAL's words.

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103 **Competing interest:** None.

104 **Financial support:** EL acknowledges funding from the Spanish Ministry of Science, Innovation &
105 Universities (MCIN/AEI/<https://doi.org/10.13039/501100011033>) under the research project
106 CNS2023-144415. The funders had no role in the decision to publish or the preparation of the
107 manuscript.

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109 **References**

110 Baggio, G. (2018). *Meaning in the Brain*. Cambridge, MA: MIT Press.

111 Dentella, V., Günther, F., & Leivada, E. (2025a). Language in vivo vs. in silico: Size matters but Larger
112 Language Models still do not comprehend language on a par with humans due to impenetrable semantic
113 reference. *PLoS One* 20(7): e0327794.

114 Dentella, V., Huang, W., Mansi, S.A., Grieve, J., & Leivada, E. (2025b). ChatGPT-generated texts show
115 authorship traits that identify them as non-human. arXiv:2508.16385.

116 Dentella, V., Günther, F., Murphy, E., Marcus, G., & Leivada, E. (2024). Testing AI on language
117 comprehension tasks reveals insensitivity to underlying meaning. *Scientific Reports* 14: 28083.

118 Diego-Simón, P.J., Chemla, E., King, J.R., & Lakretz, Y. (2025). Probing syntax in large language models:
119 successes and remaining challenges. arXiv:2508.03211.

120 Diego-Simón, P., D'Ascoli, S., Chemla, E., Lakretz, Y., & King, J.R. (2024). A polar coordinate system
121 represents syntax in large language models. In 38th Conference on Neural Information Processing Systems
122 (NeurIPS 2024), Vancouver, Canada.

123 Fodor, J., Murawski, C., & Suzuki, S. (2025). When word order matters: human brains represent sentence
124 meaning differently from large language models. *eLife* 14: RP108442.

125 Futrell, R., & Mahowald, K. (2025). How linguistics learned to stop worrying and love the language models.
126 *Behavioral and Brain Sciences*.

127 Guest, O., & Martin, A.E. (2023). On logical inference over brains, behaviour, and artificial neural networks.
128 *Computational Brain & Behavior* 6: 213–227.

129 Hadidi, N., Fegghi, E., Song, B.H., Blank, I.A., & Kao, J.C. (2025). Illusions of alignment between large
130 language models and brains emerge from fragile methods and overlooked confounds. bioRxiv
131 <https://doi.org/10.1101/2025.03.09.642245>.

132 Hunter, T. (2024). Kallini et al. (2024) do not compare impossible languages with constituency-based ones.
133 arXiv:2410.12271.

134 Jackendoff, R., & Erk, K.E. (2025). Toward a deeper lexical semantics. *Topics in Cognitive Science*
135 <https://doi.org/10.1111/tops.70013>.

136 Kobzeva, A., Arehalli, S., Linzen, T., & Kush, D. (2025). Learning filler-gap dependencies with neural
137 language models: testing island sensitivity in Norwegian and English. *Journal of Memory and Language*
138 144: 104663.

139 Krakauer, D.C., Krakauer, J.W., & Mitchell, M. (2025). Large Language Models and emergence: A complex
140 systems perspective. arXiv:2506.11135.

141 Leivada, E., Montero, R., Morosi, P., Moskvina, N., Serrano, T., Aguilar, M., & Guenther, F. (2025a). Large
142 language model probabilities cannot distinguish between possible and impossible languages.
143 arXiv:2509.15114.

144 Leivada, E., Marcus, G., Günther, F., & Murphy, E. (2025b). A sentence is worth a thousand pictures: can
 145 Large Language Models understand human language and the world behind words? *Philosophical*
 146 *Transactions of the Royal Society A* (Forthcoming).

147 Leivada, E., Murphy, E., & Marcus, G. (2023). DALL·E 2 fails to reliably capture common syntactic
 148 processes. *Social Sciences & Humanities Open* 8(1): 100648.

149 Marcolli, M., Chomsky, N., & Berwick, R.C. (2025). *Mathematical Structure of Syntactic Merge: An Algebraic*
 150 *Model for Generative Linguistics*. Cambridge, MA: MIT Press.

151 Marcus, G.F. (2001). *The Algebraic Mind: Integrating Connectionism and Cognitive Science*. Cambridge,
 152 MA: MIT Press.

153 McDermott, D. (1981). Artificial Intelligence meets natural stupidity. *ACM SIGART Bulletin* 57: 4–9.

154 Meijer, G. (2021). Neurons in the mouse brain correlate with cryptocurrency price: a cautionary tale. *Peer*
 155 *Community Journal* 1: e29.

156 Mitchell, M. (2024). The metaphors of artificial intelligence. *Science* 386(6723): eadt6140.

157 Momma, S., Richards, N., & Ferreira, V.S. (2025). Speakers encode silent structures: evidence from
 158 complementizer priming in English. *Journal of Memory and Language* 145: 104671.

159 Murphy, E. (2015). The brain dynamics of linguistic computation. *Frontiers in Psychology* 6: 1515.

160 Murphy, E. (2020). *The Oscillatory Nature of Language*. Cambridge: Cambridge University Press.

161 Murphy, E. (2024a). Predicate order and coherence in copredication. *Inquiry* 67(6): 1744–1780.

162 Murphy, E. (2024b). ROSE: A neurocomputational architecture for syntax. *Journal of Neurolinguistics* 70:
 163 101180.

164 Murphy, E. (2025). ROSE: A universal neural grammar. *Cognitive Neuroscience* 16(1-4): 49-80.

165 Murphy, E., Leivada, E., Dentella, V., Günther, F., & Marcus, G. (Forthcoming). Fundamental principles of
 166 linguistic structure are not represented by ChatGPT. *Biolinguistics*.

167 Murphy, E., de Villiers, J., & Morales, S.L. (2025). A comparative investigation of compositional syntax and
168 semantics in DALL·E and young children. *Social Sciences & Humanities Open* 11: 101332.

169 Murphy, E., Woolnough, O., Rollo, P.S., Roccaforte, Z., Segaert, K., Hagoort, P., & Tandon, N. (2022).
170 Minimal phrase composition revealed by intracranial recordings. *Journal of Neuroscience* 42(15): 3216–
171 3227.

172 Shojaee, P., Mirzadeh, I., Alizadeh, K., Horton, M., Bengio, S., & Farajtabar, M. (2025). The illusion of
173 thinking: understanding the strengths and limitations of reasoning models via the lens of problem
174 complexity. Apple Machine Learning Research report. June.

175 Slaats, S., & Martin, A.E. (2025). What’s surprising about surprisal. *Computational Brain & Behavior* 8: 233–
176 248.

177 Yang, X., Aoyama, T., Yao, Y., & Wilcox, E. (2025). Anything goes? A crosslinguistic study of (im)possible
178 language learning in LMs. *Proceedings of the 63rd Annual Meeting of the Association for Computational*
179 *Linguistics (Volume 1: Long Papers)* 26058–26077.